

TICKET SYSTEM SATISFACTION & SENTIMENT REPORT

Bootcamp Dibimbing.id – DS 35

Suci Ramadhani Fauzi | Assignment – Extra Class 5

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Tools:

1. Google Colaboratori : [Link](#)
2. Power BI : [Link](#)

Dataset:

Ticket System Review: [Link](#)

DATA UNDERSTANDING

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1462 entries, 0 to 1461
Data columns (total 10 columns):
#   Column              Non-Null Count  Dtype
---  -
0   id_survey           1462 non-null   object
1   date_of_survey      1462 non-null   object
2   ticket_system       1462 non-null   object
3   overall_rating      787 non-null    float64
4   customer_service    787 non-null    float64
5   features            787 non-null    float64
6   value_for_money     787 non-null    float64
7   ease_of_use         787 non-null    float64
8   likelihood_to_recomm 787 non-null    float64
9   overall_text        787 non-null    object
dtypes: float64(6), object(4)
memory usage: 114.3+ KB
```

- Dataset: Ticket system reviews (Oct–Dec 2024).
- Ukuran data
 - 1,462 entri (tiket) total.
 - 787 entri mengisi survey (54%), 675 tidak mengisi (46%).
- Fitur utama
 - id_survey, date_of_survey, ticket_system
 - overall_rating, customer_service, features, value_for_money, ease_of_use, likelihood_to_recommend, overall_text
- Tujuan: mengukur CSAT, CES (Ease of Use), NPS, dan menganalisis sentimen teks ulasan.

METODE PENGOLAHAN DATA

Preprocessing data terstruktur

- Filter hanya responden yang mengisi rating/teks (drop NA pada overall_rating/overall_text).
- Normalisasi tipe data (tanggal ke datetime, rating 1–5).
- Standarisasi kategori ticket_system.

Metode analisis

- $CSAT\% = \frac{SUM(rating)}{(COUNT(responden) \times 5)}$.
- $CES\% = \text{rata-rata ease_of_use} / 5 \times 100$.
- $NPS = \%Promoters(9-10) - \%Detractors(0-6)$ dari likelihood_to_recommend.
- Sentimen: pendekatan lexicon-based (mis. VADER) untuk ulasan pendek.

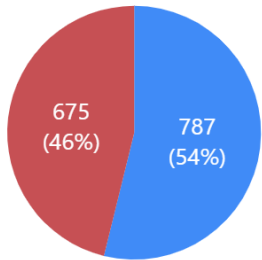
Preprocessing teks

- Lowercase, hapus tanda baca/angka, hapus stopwords, lemmatization.
- Deteksi bahasa & jaga kata kunci produk/brand.

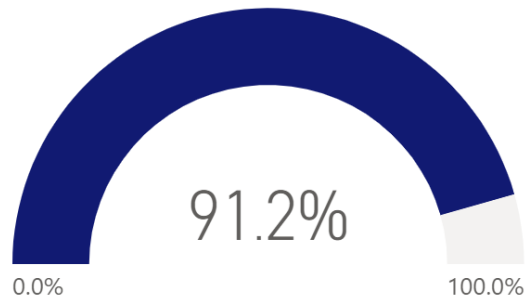
RINGKASAN HASIL KPI

Survey Response

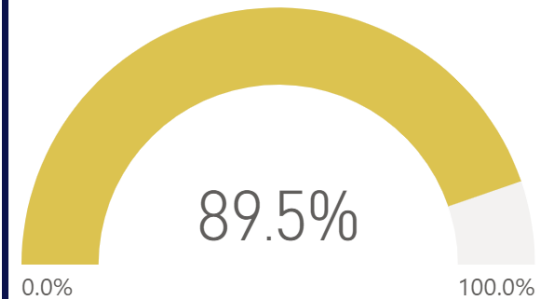
Response ● Responded ● Not Responded



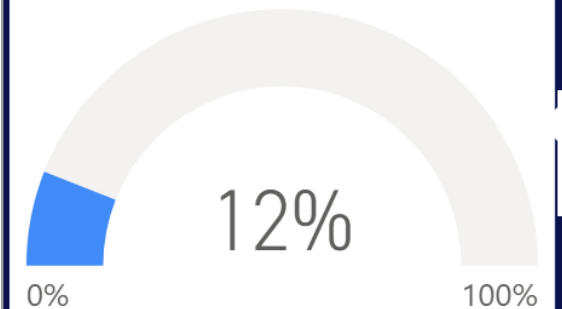
CSAT



CES Ease of Use



NPS Score



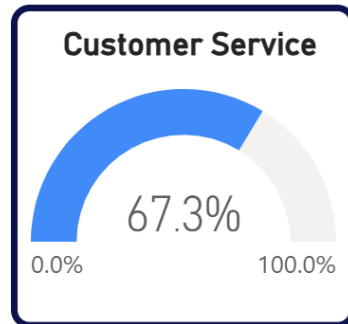
Survey Response Rate:
54%
(787/1,462).

CSAT: 91.2%
(sangat tinggi &
konsisten).

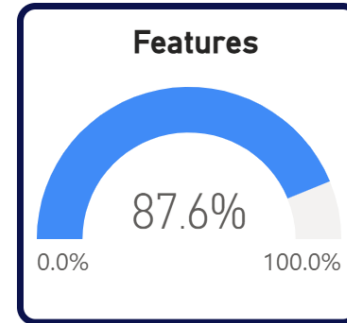
CES (Ease of Use):
89.5%
(tinggi, UX cukup
mudah).

NPS: 12%
(masih moderat; ada
porsi detractors yang
perlu ditangani).

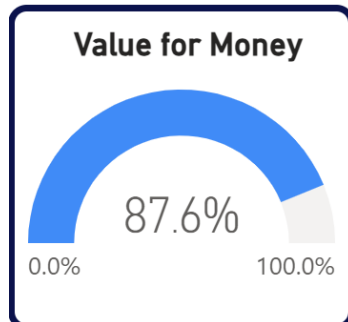
DIMENSI KEPUASAN



Customer Service:
67.3%
(terendah → prioritas peningkatan).



Features: 87.6%
(kuat).

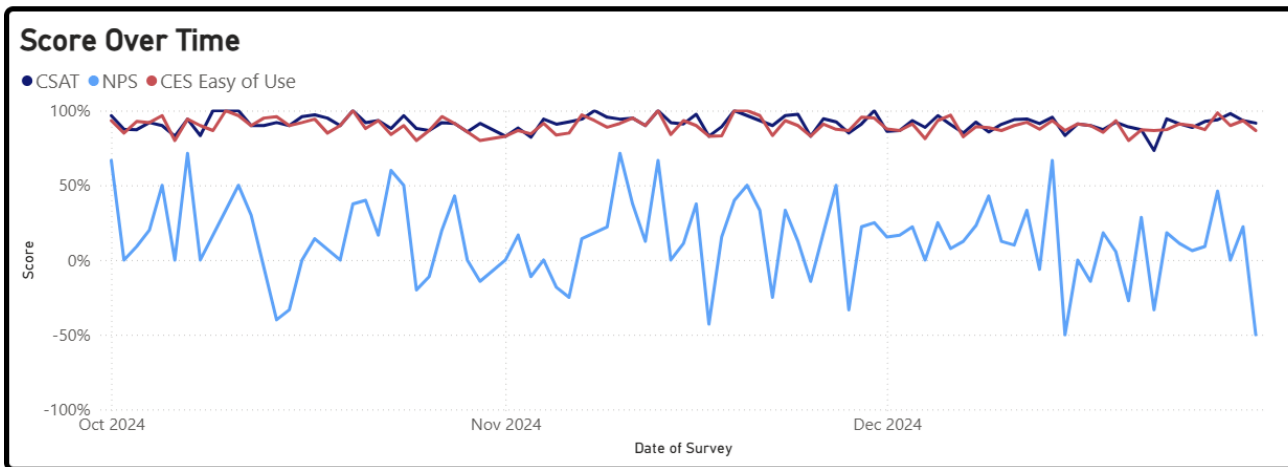


Value for Money:
87.6% (baik).

PERBANDINGAN ANTAR TIKET SYSTEM & TREND SKOR DARI WAKTU KE WAKTU

Ticket Systems	Responden	CAST	NPS	CES Easy of Use
Freshdesk	355	92.1%	24%	92.8%
Jira Service Management	144	91.7%	8%	87.7%
otrs	36	91.1%	22%	86.7%
ServiceNow	54	90.4%	15%	84.4%
Zendesk	435	90.8%	6%	88.0%
Zoho Desk	438	90.8%	7%	89.5%
Total	1462	91.2%	12%	89.5%

- Responden terbesar: Zendesk (435), Zoho Desk (438), Freshdesk (355).
- CSAT relatif seragam ~90–92% di seluruh vendor.
- NPS bervariasi: Freshdesk 24% (tertinggi), ServiceNow 15%, OTRS 22%, Zendesk 6–7% (terendah).
- CES tertinggi: Freshdesk 92.8%; terendah: ServiceNow 84.4%.

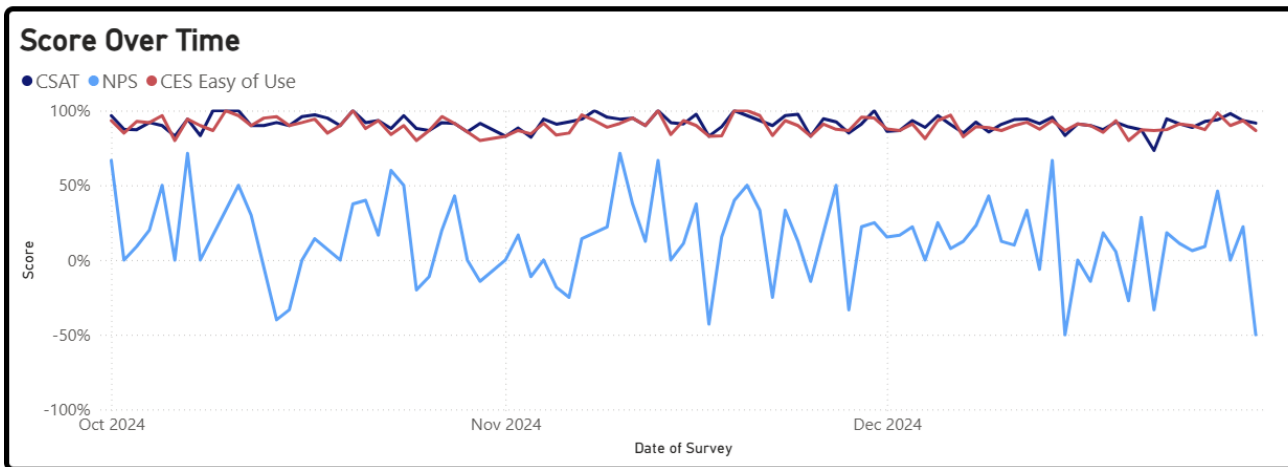


- Skor CSAT & CES relatif stabil di level tinggi; NPS berfluktuasi namun cenderung positif.
- Perlu investigasi pada titik-titik penurunan harian untuk menemukan penyebab (release, outage, backlog).

PERBANDINGAN ANTAR TIKET SYSTEM & TREK SKOR DARI WAKTU KE WAKTU

Ticket Systems	Responden	CAST	NPS	CES Easy of Use
Freshdesk	355	92.1%	24%	92.8%
Jira Service Management	144	91.7%	8%	87.7%
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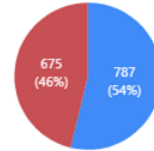
DASHBOARD

Power BI : [Link](#)

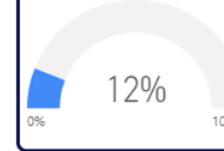
TICKET SYSTEM SATISFACTION REPORT

Survey Response

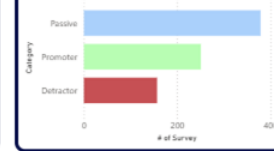
Response ● Responded ● Not Responded



NPS Score

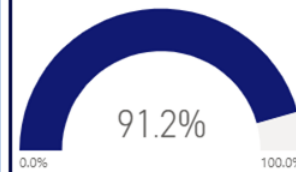


NPS Category

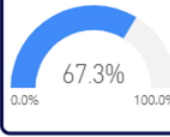


Customer Satisfaction

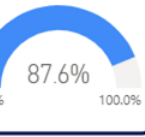
CSAT



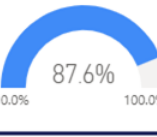
Customer Service



Features

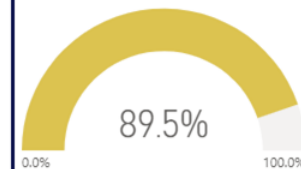


Value for Money



Customer Effort

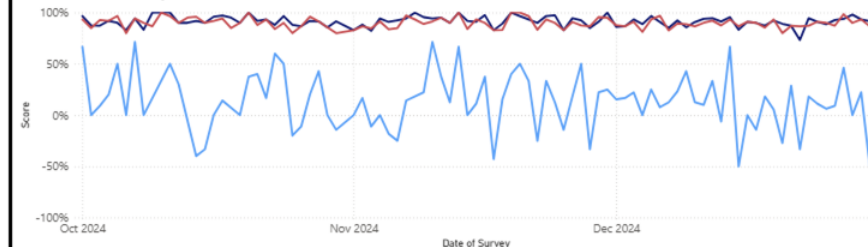
CES Ease of Use



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Score Over Time

Score ● CSAT ● NPS ● CES Easy of Use



SENTIMENT INSIGHTS (WORD CLOUD)



- **Positif:** “easy, use, support, tool, ticket” → sistem mudah dipakai, dukungan cukup membantu.
- **Negatif:** “need, reporting/analytics, service, time, ui, access” → butuh laporan lebih baik, layanan lambat, ada friksi UX (izin/navigasi).

INSIGHT UTAMA

- Kepuasan keseluruhan (CSAT 91.2%) sangat baik; pengalaman penggunaan (CES 89.5%) juga tinggi.
- Faktor yang paling lemah adalah 'Customer Service' (67.3%): waktu respons, kualitas resolusi, atau empati agen perlu dievaluasi.
- NPS 12% menunjukkan masih ada kantong detractors dengan isu berulang (kata kunci negatif: "service", "zendesk", "reporting", "need", "problem").
- Kata kunci positif menekankan kemudahan (easy), dukungan (support), dan value tool (ticket, customer, tool).

REKOMENDASI STRATEGIS

- Perbaiki Customer Service (prioritas)
 - Penguatan SLA: target response & resolution time per kategori prioritas.
 - Coaching agen: script, empathy, root-cause handling, dan knowledge-base update.
 - Quality assurance: sampling chat/email untuk evaluasi berkala.
- Optimasi UX untuk dongkrak CES
 - Perbaiki alur kerja tiket: template, auto-suggest solusi, dan navigasi yang lebih ringkas.
 - Sediakan self-service portal & artikel KB dengan pencarian yang lebih baik.
- Kelola Detractors (tingkatkan NPS)
 - Program 'close-the-loop': hubungi detractors, klasifikasikan isu, dan tindak lanjut dalam 24–48 jam.
 - Analisis topik otomatis dari ulasan negatif untuk prioritas backlog product/ops.
- Pemantauan Berkelanjutan
 - Bangun dashboard operasional per sistem (SLA, backlog, first-contact resolution).
 - A/B testing terhadap perubahan proses/fitur untuk melihat dampak pada CSAT/CES/NPS.

LAMPIRAN – DEFINISI & FORMULA

- $\text{CSAT\%} = \text{SUM}(\text{rating}) / (\text{COUNT}(\text{responden}) \times \text{rating_maks}) \times 100$ (rating_maks = 5).
- $\text{CES\%} = \text{AVERAGE}(\text{ease_of_use}) / 5 \times 100$.
- $\text{NPS} = \%\text{Promoters (9–10)} - \%\text{Detractors (0–6)}$ dari likelihood_to_recommend.
- Sentimen: klasifikasi positif/negatif berbasis leksikon; dapat ditingkatkan dengan model supervised TF-IDF + Logistic Regression atau transformer.

TERIMA KASIH